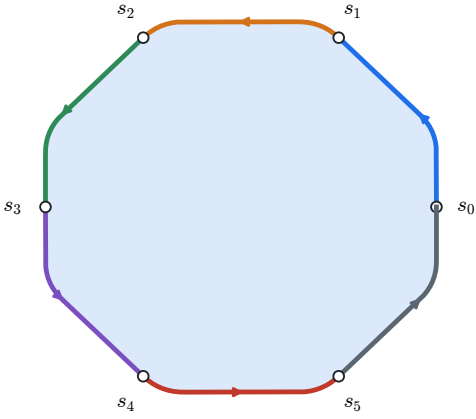
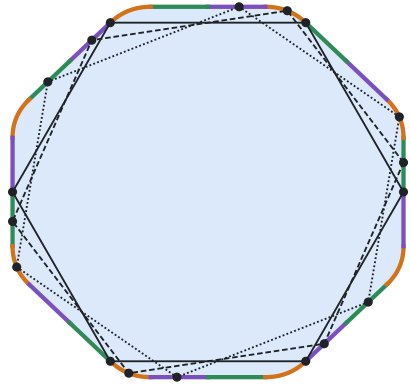


(a) the arcs $g(t)s_j$, $0 \le t \le t_{\mathrm{f}}$



(b) contact hexagons at $t = 0,\ t_{\mathrm{sw}}/2,\ 3t_{\mathrm{sw}}/2$

